

Statement of Purpose

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As an undergraduate research assistant in a machine learning lab, I worked on model optimization. The work was technically interesting, but I wanted research that helps people more directly. When I joined the Interactive Data Computing Laboratory and started building visualization tools, I found that connection. Seeing domain experts actually use what I built showed me the direct impact of HCI research. Watching them change how they work, not just once but across multiple projects, convinced me to pursue this direction.

Domain experts understand their data, but they often lack tools to visually explore it. A machine learning researcher knows which hyperparameters matter, yet existing dashboards show experiment logs without connecting interventions to outcomes. A software testing researcher cannot see which parameters affect branch coverage, resorting to spreadsheet averages. A pediatrician may follow a patient for years, but fragmented records make it difficult to discern longitudinal patterns. I build **visual analytics systems** for these experts. Through three projects in machine learning, software testing, and pediatric medicine, I found that effective visualization must adapt to how experts already think. At the University of Washington, I will develop authoring frameworks that enable domain experts to build visual exploration tools for their own workflows.

I have designed and evaluated **three interactive visual analytics systems**:

I developed **HyPockeTuner**, an interactive mobile system for managing hyperparameter optimization (HPO) in machine learning on smartphones. Eight ML researchers told me that existing tools, such as Weights & Biases, (1) lack adequate visual support for understanding experiment history and user interactions, (2) are designed for desktop environments with little to no mobile support, and (3) provide limited control for modifying search spaces during experiments. To address these issues, I built EventCrumb, a mobile event sequence visualization that integrates experiment progress, user interventions, and results on a unified timeline. In a lab study with 12 participants, EventCrumb achieved 96.9% accuracy for understanding experiment context and scored 82.3 on the System Usability Scale. Two researchers used it for five days and were able to conduct experiments during lunch breaks or while commuting, rather than staying late at the office. The unified timeline made mobile monitoring practical. This project showed me that human intervention can complement automation when intermediate results are visible and actionable.

In my second project, I built **Symetra**, a visual analytics system for tuning symbolic execution engine parameters in software testing. Over ten months, I collaborated with two domain experts to build a system that analyzes thousands of parameter configurations. While existing tools for HPO focus on single metrics and handle several hyperparameters, software testing requires analyzing tens to hundreds of parameters and branch coverage vectors. I designed two overviews: one for analyzing the impacts of parameters on branch coverage, and another for identifying trials with similar or complementary coverage patterns. Users could group trials, perform comparative analyses, and identify complementary parameter combinations that cover different branches. When experts used the system, they found combinations of parameters they had never known to work together. In a Human-in-the-Loop use case, this approach achieved 17.5% coverage improvement and eliminated failed runs. Building Symetra from scratch showed that each domain requires its own visualization design. This raised a question: if every domain requires custom tools, how can we help

experts build their own?

I refined HyPockeTuner and Symetra through three revision cycles each since April 2023. With each iteration, I addressed reviewer feedback by adding contributions and conducting additional user studies and expert interviews. Through this process, I learned to connect the research question, the system, and the evaluation into one coherent narrative.

The last project began when pediatricians at Seoul National University Hospital asked for help visualizing pediatric chronic disease records scattered across separate systems. To address this, I designed **PediaViz**, a system that brings diagnoses, prescriptions, lab results, and growth trajectories into a single dashboard. As the project progressed through several prototype cycles, the scope expanded to include clinical notes alongside structured data. In our monthly discussions, I realized that pediatricians reason in terms of long-term histories rather than isolated visits. This led me to redesign the system around longitudinal views instead of short-term summaries. Working with pediatricians confirmed what I observed across all three projects. Effective visualization adapts to existing expert workflows, not the other way around.

Beyond research, my experience as Student Council Vice President at the College of Computing taught me to identify structural problems and work with collaborators. When I joined the council, poorly defined roles forced members to handle marketing, event planning, and communications simultaneously, causing some members to leave. After being elected vice president, I collaborated with council department directors to clarify responsibilities and transition meetings from online to in-person formats. I also resumed student events suspended during COVID-19, including orientation and Homecoming. These experiences shaped my approach to research: first, understanding how people work, then building systems that support them.

My doctoral research asks: **How can we design tools that let domain experts build visual analytics interfaces without relying on developers?** Building three systems from scratch taught me that each domain needs its own visualization, but experts share similar analytical practices: tracking changes, comparing alternatives, and iterating on intermediate results. I will study how these patterns can inform reusable components for authoring domain-specific visual analytics tools.

The Interactive Data Lab at the University of Washington combines declarative visualization design with scalable systems research, both of which are essential to my goal of building authoring frameworks for domain experts. **Professor Jeffrey Heer**'s work on visualization grammars, including Vega-Lite, demonstrates how high-level specifications can make visualization authoring accessible. I want to explore how similar abstractions could help domain experts define visual analytics workflows, turning their analytical patterns into reusable components. **Professor Leilani Battle** builds systems that keep visualizations responsive as data scales. My work faced a similar challenge. HyPockeTuner needed to provide immediate feedback as users explored hundreds of hyperparameter configurations, and her research on scalable interactive systems would help me build authoring frameworks that remain interactive at scale.

My long-term goal is to lead an academic lab that builds **open-source authoring tools for domain experts**. Over two years as a teaching assistant for HCI courses, I found that guiding students through design challenges motivated me as much as my own research. Helping about 130 students scope design projects and run evaluations taught me to give useful feedback. I want to bring this to mentoring graduate researchers. At the University of Washington, I will develop authoring frameworks that enable domain experts to build visualization tools tailored to their own workflows.